

Consumer acceptance of vegetarian sweet potato products intended for space missions.

Sweet potato is one of the crops selected for NASA's Advanced Life Support Program for potential long-duration lunar/Mars missions. This article presents recipes of products made from sweet potato and determines the consumer acceptability of products containing from 6% to 20% sweet potato on a dry weight basis. These products were developed for use in nutritious and palatable meals for future space explorers. Sensory evaluation (appearance/color, aroma, texture, flavor/taste, and overall acceptability) studies were conducted to determine the consumer acceptability of vegetarian products made with sweet potato using panelists at NASA/Johnson Space Center in Houston, TX. None of these products including the controls, contained any ingredient of animal origin with the exception of sweet potato pie. A 9-point hedonic scale (9 being like extremely and 1 being dislike extremely) was used to evaluate 10 products and compare them to similar commercially available products used as controls. The products tested were pancakes, waffles, tortillas, bread, pie, pound cake, pasta, vegetable patties, doughnuts, and pretzels. All of the products were either liked moderately or liked slightly with the exception of the sweet potato vegetable patties, which were neither liked nor disliked. Mean comparisons of sensory scores of sweet potato recipes and their controls were accomplished by using the Student t-test. Because of their nutritional adequacy and consumer acceptability, these products are being recommended to NASA's Advanced Life Support Program for inclusion in a vegetarian menu plan designed for lunar/Mars space missions.